
BAR: Budgeted Actor Refinement Broadens the Stability Envelope of Offline TD3+BC

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Abstract

TD3+BC uses one actor branch both to construct bootstrapped critic targets and to produce the deployed policy, tying target-policy movement to deployment reach. We introduce budgeted actor refinement (BAR), a persistent TD3+BC actor chain that splits a nominal improvement budget across local updates: the first-hop target copy supplies TD actions, while the final hop is deployed. The squared action anchor has a deterministic Wasserstein-2 proximal interpretation. We derive an inexact finite-step margin relative to the learned critic, an explicit critic-error term, a first-hop target-perturbation bound, and a fixed- K local expansion. Across nine D4RL locomotion tasks and fourteen budgets, four proximal actors raise the high-budget mean from 25.97 to 63.61 and reduce low-return cells from 35/63 to 9/63. The $K = 4$ minus $K = 3$ mean gain is 10.83, but its median is only .32, locating the improvement in failure rescue. A linearized realization improves in the same direction. Three-hop BAR also lowers directly measured target-policy displacement in 88/90 matched setting comparisons while lowering the final-policy displacement proxy in only 36/90. These results support an end-to-end stability-envelope gain for the implemented refinement bundle. They do not isolate routing, re-centering, or compute.

1 INTRODUCTION

Offline reinforcement learning (RL) seeks a policy from a fixed dataset without additional environment interaction. Its central difficulty is not merely to select a high-value action, but to determine how far policy improvement can proceed before the learned critic becomes unreliable. TD3+BC regularizes a deterministic actor toward dataset actions while using a Polyak copy of that actor in bootstrapped critic targets (Fujimoto and Gu, 2021). The same policy branch therefore determines both the action shift entering critic training and the reach of the evaluated policy. Aggressive improvement can move target actions into regions where critic error is amplified, whereas conservative improvement can leave useful policy improvement unrealized.

We study this coupling through *budgeted actor refinement* (BAR), a re-centered actor architecture for TD3+BC. BAR represents a nominal improvement budget with K sequential actor updates. The first actor is anchored to dataset actions, and each later actor is anchored to the stop-gradient output of its predecessor. A target copy of the first refined actor supplies the next actions in TD targets, while the final actor is used for evaluation. The architecture thus assigns critic-target construction and deployment to distinct policy branches without changing the critic update rule.

Let D_{critic} denote the squared displacement of the critic-coupled first-hop branch and let D_{final} denote that of the evaluated final policy. A one-hop actor identifies these quantities; BAR assigns them to μ_1 and μ_K , respectively. This separation can increase final-policy reach at comparable first-hop movement or contract both movements near an unstable training outcome. It also introduces a countervailing mismatch because the critic target follows the μ_1 branch while evaluation follows μ_K . The method therefore exposes a design axis between target-policy displacement and deployment reach rather than guaranteeing that either quantity is independently optimal.

For deterministic actors, the squared action anchor in TD3+BC is a Wasserstein-2 movement cost in the Dirac-policy limit. Under a fixed critic, inactive action projection, accurate subproblem solutions, and small *realized* steps, explicit and implicit updates are first-order discretizations of the same Wasserstein gradient flow. At finite steps, a proximal comparison yields an improvement margin only relative to the learned critic. At large nominal budgets, projection, neural actor fitting, critic drift, saturation, and the seeded optimization trajectory determine the observed end-to-end outcome. We keep these local, finite-step, and coupled regimes distinct throughout the analysis.

We evaluate this formulation with a complete fixed-budget phase diagram over nine D4RL locomotion tasks, fourteen budgets, and two seeds. Four proximal actors reduce large-budget low-return cells from 35/63 for TD3+BC to 9/63 and raise the high-budget mean from 25.97 to 63.61. The mean improvement from $K = 3$ to $K = 4$ is large, whereas the median is nearly zero, showing lower-tail rescue rather than a uniform score shift. Three scale-matched projected-linearized actors reduce failures to 24/63, so the direction is not unique to the proximal loss. The one-hop phase diagram reproduces on two further seeds. Direct next-state diagnostics show that three-hop BAR lowers deterministic target-policy displacement in 88/90 matched setting comparisons with TD3+BC, while its current-state final-policy displacement proxy is lower in only 36/90. Low-return runs additionally exhibit orders-of-magnitude increases in TD-error tails, larger realized movements, and critic gains that reverse sign under a controlled common critic.

Our contributions are as follows.

- We formulate BAR for TD3+BC as a re-centered actor composition that assigns target-policy displacement and final-policy reach to distinct branches.
- We give a Wasserstein proximal interpretation, derive an inexact finite-step margin with an explicit critic-error term, bound first-hop TD-target perturbation, and separate these statements from the fixed-critic small-step flow limit.
- Through complete proximal sweeps up to $K = 4$, a complete linearized $K = 3$ sweep, direct target-policy measurements, critic-failure diagnostics, and targeted controls, we show that the tested refinement bundles broaden the empirical large-budget stability envelope primarily by rescuing lower-tail failures.

2 RELATED WORK

Behavior-regularized offline actor-critic learning. Offline actor-critic methods must control value extrapolation when a learned policy selects actions outside the fixed dataset. Prior work studies this failure through constrained policy improvement, conservative value learning, and restrictions on actions used in Bellman backups (Fujimoto et al., 2019; Kumar et al., 2019). TD3+BC provides a particularly minimal control: a deterministic actor maximizes a normalized critic while a squared cloning term anchors its actions to the dataset (Fujimoto and Gu, 2021). We keep this critic update and actor family fixed and study which refined actor branch supplies the TD target.

Policy geometry and proximal refinement. Trust-region policy optimization is commonly derived from information geometry (Amari and Nagaoka, 2000; Schulman et al., 2015). Optimal-transport geometry gives a different notion of movement: policy optimization can be expressed as a Wasserstein gradient flow, with related natural-gradient and trust-region constructions for continuous control (Zhang et al., 2018; Moskovitz et al., 2020; Terpin et al., 2022; Song et al., 2023). For deterministic policies, the statewise Wasserstein-2 cost reduces to squared action displacement. We use that identity to interpret the TD3+BC anchor as a proximal movement cost, while separating the finite-step variational statement from its small-step flow limit.

Target networks and actor-critic coupling. Bootstrapping error can feed critic extrapolation back into actor updates (Fujimoto et al., 2019; Kumar et al., 2019). Polyak averaging delays this feedback but ordinarily copies the same actor branch that will be deployed. Target-deployment routed refinement makes a narrower architectural change: the target copy follows the first local refinement, whereas evaluation uses the final refinement. Our target-perturbation bound and direct displacement audit therefore concern branch assignment inside TD3+BC rather than a new critic objective or a general claim about offline policy extraction.

3 PRELIMINARIES

We consider a discounted Markov decision process $\langle \mathcal{S}, \mathcal{A}, P, r, \gamma \rangle$ with continuous actions and a fixed offline dataset $\mathcal{D} = \{(s_i, a_i, r_i, s'_i)\}_{i=1}^N$. A clipped double critic is trained from targets

$$y_\rho = r + \gamma \min_j Q_j^-(s', \rho(s')), \quad (1)$$

where ρ is a target policy. When $\rho(s')$ leaves dataset support, critic error enters the regression target and can be amplified by later actor updates (Fujimoto et al., 2019; Kumar et al., 2019).

For a deterministic actor μ_θ , normalized TD3+BC minimizes

$$\mathcal{L}_{\text{actor}}(\theta) = -\alpha \mathbb{E}_s \bar{Q}(s, \mu_\theta(s)) + \mathbb{E}_{(s,a) \sim \mathcal{D}} \frac{\|\mu_\theta(s) - a\|_2^2}{d}, \quad (2)$$

where $\bar{Q} = \hat{Q}/C$ and $C > 0$ is a stop-gradient critic scale (Fujimoto and Gu, 2021). Ordinarily, a Polyak copy of this actor enters Eq. (1), while the online actor is deployed. Target-policy movement and deployment reach therefore follow the same branch.

A metric proximal update with step $h > 0$ takes the form

$$\pi^+ \in \arg \min_{\pi} \left\{ \hat{\mathcal{E}}(\pi) + \frac{1}{2h} d_{\mathcal{M}}^2(\pi, \pi^{\text{ref}}) \right\}. \quad (3)$$

This minimizing-movement construction is associated with metric gradient flows (Ambrosio et al., 2005; Otto, 2001). For deterministic state-conditioned policies,

$$W_2^2(\delta_{\mu(s)}, \delta_{\nu(s)}) = \|\mu(s) - \nu(s)\|_2^2. \quad (4)$$

Thus the TD3+BC anchor is a Wasserstein-2 movement cost up to fixed scaling. We distinguish its finite-step variational meaning from the small-step, fixed-critic gradient-flow limit and from large-budget coupled training.

4 BUDGETED ACTOR REFINEMENT

At each delayed actor update, BAR maintains K independently initialized, persistent deterministic actors with separate optimizer states. The first actor takes one TD3+BC update toward the sampled dataset actions; each later actor takes one update toward the stop-gradient output of its predecessor on the same minibatch. The online critic is fixed within this chain; we denote its first Q-function by $\hat{Q}$. A Polyak copy of the first actor supplies TD-target actions, and the final actor is evaluated. We first derive the proximal operator underlying these updates and then formalize their composition.

4.1 Offline actor updates as proximal improvement

Let $\rho_{\mathcal{D}}$ be the empirical state distribution and consider a deterministic actor $\mu : \mathcal{S} \rightarrow \mathcal{A} \subset \mathbb{R}^d$. At one training iteration, the critic parameters are held fixed while the

actor chain is updated. We write the current critic as $\hat{Q}$ and normalize its global scale by

$$\bar{Q}_k(s, a) = \frac{\hat{Q}(s, a)}{C_k}, \quad (5)$$

$$C_k = \text{sg} \left(\mathbb{E}_s |\hat{Q}(s, \tilde{\mu}_k(s))| + \epsilon \right).$$

where $\tilde{\mu}_k$ is the actor before hop k and sg denotes stop-gradient. A fixed positive C only rescales gradient-flow time; recomputing C_k changes the finite-step schedule. Our finite-step statements condition on the realized scales.

For deterministic policies and a state distribution ρ , define the action-averaged Wasserstein metric

$$\mathbf{d}_{d,\rho}^2(\mu, \nu) := \mathbb{E}_{s \sim \rho} \left[\frac{1}{d} \|\mu(s) - \nu(s)\|_2^2 \right]. \quad (6)$$

It is d^{-1} times the state-averaged squared W_2 distance between state-conditioned Dirac policies. We abbreviate $\mathbf{d}_d = \mathbf{d}_{d,\rho_{\mathcal{D}}}$ on the actor-training state distribution. The population counterpart of squared behavior cloning also admits a deterministic reference. Let P_B be the data-generating behavior distribution, let ρ_B be its state marginal, and define $b(s) = \mathbb{E}_{P_B}[a | s]$. Then

$$\mathbb{E}_{(s,a) \sim P_B} \frac{1}{d} \|\mu(s) - a\|_2^2 = \mathbf{d}_{d,\rho_B}^2(\mu, b) + \text{const}, \quad (7)$$

where the constant is the conditional action variance and does not depend on μ . The empirical actor loss is a Monte Carlo estimate of the left-hand side, so samplewise squared anchoring targets the conditional-mean behavior map at the population level.

Given a reference policy ν and step $h > 0$, define the proximal refinement operator by

$$\mathcal{P}_h^{\hat{Q}}(\nu) \in \arg \min_{\mu \in \mathcal{M}} \left\{ -\mathbb{E}_s \bar{Q}_k(s, \mu(s)) + \frac{1}{2h} \mathbf{d}_d^2(\mu, \nu) \right\}. \quad (8)$$

Multiplying by $2h$ gives the practical actor objective

$$\mathcal{L}_{h,k}(\mu | \nu) = -2h \mathbb{E}_s \bar{Q}_k(s, \mu(s)) + \mathbf{d}_d^2(\mu, \nu). \quad (9)$$

Proposition 1 (Behavior anchoring is Wasserstein proximal improvement). *For deterministic policies, the normalized anchored objective*

$$-\alpha \mathbb{E}_s \bar{Q}(s, \mu(s)) + \mathbf{d}_d^2(\mu, \nu)$$

is Eq. (8), up to multiplication by a positive constant, with $h = \alpha/2$. Equivalently, the raw-critic objective $-\lambda \mathbb{E}_s \hat{Q}(s, \mu(s)) + \mathbf{d}_d^2(\mu, \nu)$ has $h = \lambda C/2$ when C is fixed. The population counterpart of squared dataset anchoring uses $\nu = b$ by Eq. (7).

The proposition identifies an update operator relative to the *estimated* critic; true-return improvement lies outside the scope of this certificate.

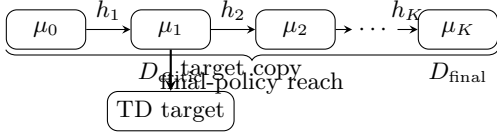


Figure 1: BAR re-centers each actor update. A Polyak copy of μ_1 supplies TD-target actions, while later actors increase or reshape final-policy reach without entering the target policy.

4.2 Re-centered composition and actor-critic dataflow

For population analysis, let $\mu_0 := b$ denote the conditional-mean behavior map and let $h_{1:K}$ be positive step sizes. The idealized BAR chain is

$$\mu_k \in \mathcal{P}_{h_k}^{\widehat{Q}}(\text{sg } \mu_{k-1}), \quad k = 1, \dots, K. \quad (10)$$

The implementation approximates each subproblem with one optimizer update. Its first actor μ_1 uses sampled dataset actions as its reference; later actors $\mu_2, \dots, \mu_K$ use the preceding online actor under the same current critic. Only the Polyak copy of μ_1 enters the TD target, while μ_K supplies the evaluated action.

Our fixed-budget schedule sets $h_k = T/K$ and holds the total nominal budget T fixed. Increasing K jointly changes local step size, re-centering frequency, actor compute, and the target/deployment dataflow; the experiment therefore estimates the effect of this implemented refinement bundle.

On the next-state distribution $\rho'_{\mathcal{D}}$ used by the TD target, define the action-averaged population displacements

$$\begin{aligned} D_{\text{critic}} &:= d_{d, \rho'_{\mathcal{D}}}^2(\mu_1, \mu_0), \\ D_{\text{final}} &:= d_{d, \rho'_{\mathcal{D}}}^2(\mu_K, \mu_0). \end{aligned} \quad (11)$$

A one-step actor has $D_{\text{critic}} = D_{\text{final}}$; BAR assigns the two population quantities to distinct policy branches. On fixed validation batches, we measure the sample-anchored online branches and the deterministic next-state Polyak target branch:

$$\begin{aligned} \widehat{D}_{\text{critic}} &= \mathbb{E}_i \|\mu_1(s_i) - a_i\|_2^2/d, \\ \widehat{D}_{\text{final}} &= \mathbb{E}_i \|\mu_K(s_i) - a_i\|_2^2/d, \\ \widehat{D}_{\text{target}} &= \mathbb{E}_i \|\mu_1^-(s'_i) - a_{i+1}\|_2^2/d. \end{aligned} \quad (12)$$

The direct audit retains only bootstrapped transition pairs; a smoothed version uses common TD3 target-noise draws. Sample anchoring adds a common conditional-action-variance floor in population differences. These are movement measurements, not critic-reliability estimates.

4.3 Finite-step statements

The following statements do not require a small-step Taylor expansion.

Proposition 2 (Inexact estimated-energy margin). *Suppose μ_{k-1} is feasible for the k th actor class and the computed actor satisfies*

$$\mathcal{L}_{h_k, k}(\mu_k \mid \mu_{k-1}) \leq \mathcal{L}_{h_k, k}(\mu_{k-1} \mid \mu_{k-1}) + \varepsilon_k.$$

Then

$$\begin{aligned} m_k^{\widehat{Q}} &:= \frac{2h_k}{C_k} \mathbb{E}_s [\widehat{Q}(s, \mu_k(s)) - \widehat{Q}(s, \mu_{k-1}(s))] \\ &\quad - d_d^2(\mu_k, \mu_{k-1}) \geq -\varepsilon_k. \end{aligned} \quad (13)$$

For the first sample-anchored hop, a statewise dataset action is not generally a feasible deterministic-policy comparator; the population reference in Eq. (7) or a behavior actor is required for a literal certificate. For re-centered hops, the preceding actor is feasible by construction.

The margin is relative to $\widehat{Q}$. Let a trusted objective be $Q^\dagger$ and write $\widehat{Q} = Q^\dagger + e$. Then

$$\begin{aligned} m_k^{Q^\dagger} &= m_k^{\widehat{Q}} - \frac{2h_k}{C_k} \mathbb{E}_s [e(s, \mu_k(s)) \\ &\quad - e(s, \mu_{k-1}(s))]. \end{aligned} \quad (14)$$

Thus estimated-energy descent transfers to $Q^\dagger$ only when critic error is controlled along the realized path. Equation (13) certifies the learned objective rather than monotone true return.

Proposition 3 (First-hop TD-target perturbation bound). *For clarity, consider a noiseless TD target using μ_1 , $y_{\mu_1} = r + \gamma \min_j Q_j^-(s', \mu_1(s'))$, and suppose each target critic $Q_j^-(s', \cdot)$ is $L(s')$ -Lipschitz. For any reference μ_0 ,*

$$\begin{aligned} \|y_{\mu_1} - y_{\mu_0}\|_{L_2(\mathcal{D})} &\leq \gamma \left(\mathbb{E}_{s'} L(s')^2 \right. \\ &\quad \cdot \|\mu_1(s') - \mu_0(s')\|_2^2 \Big)^{1/2}. \end{aligned} \quad (15)$$

If $L(s') \leq \bar{L}$, the right-hand side is at most $\gamma \bar{L} \sqrt{d D_{\text{critic}}}$. Conditional on μ_1 , later actors do not enter this one-step target perturbation.

Proposition 3 bounds a one-step value-target difference, not the stability of the learned critic. Its architectural content is that later refinement can alter D_{final} without directly entering this bound. With a Polyak target network, μ_1 is replaced by its target copy. A same-noise comparison under TD3 target smoothing obeys the same bound because action clipping is non-expansive. The current-state $\widehat{D}_{\text{critic}}$ omits Polyak lag

and target noise, whereas $\hat{D}_{\text{target}}$ measures the Polyak actor directly and has a smoothed counterpart. Neither displacement measures the action Lipschitz factor or the resulting change in critic value.

The separation introduces a complementary evaluation-policy mismatch: critic targets follow the μ_1 branch, whereas evaluation follows μ_K . Increasing D_{final} at fixed D_{critic} can therefore make the critic less representative of the evaluated policy. We interpret BAR as exposing a measurable target/deployment trade-off; its empirical return captures the net effect of target movement, evaluation mismatch, actor compute, and the coupled optimization trajectory.

4.4 Local Wasserstein-flow limit

The gradient-flow interpretation is a local organizing principle. Fix a critic and an interior action a_0 . Let $g = \nabla_a \hat{Q}(a_0)$, $H = \nabla_a^2 \hat{Q}(a_0)$, and $\beta = dh/C$. Before projection, the scale-matched linearized and exact statewise proximal updates satisfy

$$\Delta a_{\text{lin}} = \beta g, \quad \Delta a_{\text{prox}} = \beta \nabla_a \hat{Q}(a_0 + \Delta a_{\text{prox}}). \quad (16)$$

When $\beta \|H\| \ll 1$, the latter yields

$$\Delta a_{\text{prox}} = (I - \beta H)^{-1} \beta g = \beta g + \beta^2 H g + O(\beta^3), \quad (17)$$

and both updates are first-order discretizations of $\dot{a} = d \nabla_a \hat{Q}(a)$. With fixed C this is a constant time-rescaling of raw-critic gradient ascent; with a positive policy-dependent global C it follows the same direction under an adaptive time parameterization. This expansion additionally requires small realized displacement, inactive projection, and a sufficiently accurate actor solve. At large budgets, our experiments diagnose the resulting nonlocal actor-critic regime instead of extrapolating Eq. (17). The end-to-end return sweep, even at small nominal T , is not a frozen-critic action-level consistency test of this limit.

Proposition 4 (Local fixed-budget composition). *Let a smooth one-step update in local coordinates satisfy $\Psi_h(x) = x + hf(x) + h^2p(x) + O(h^3)$. For fixed K and $T \rightarrow 0$,*

$$\begin{aligned} \Psi_{T/K}^K(x_0) &= x_0 + T f_0 \\ &+ T^2 \left[\frac{K-1}{2K} J_0 f_0 + \frac{1}{K} p_0 \right] + O(T^3). \end{aligned} \quad (18)$$

where $f_0 = f(x_0)$ and $J_0 = Df(x_0)$. Coordinate explicit Euler has $p_0 = 0$, whereas exact backward Euler has $p_0 = J_0 f_0$. Hence both are first-order methods; for each fixed K , the $T \rightarrow 0$ expansion has a leading flow-bias coefficient proportional to $1/K$.

Proposition 4 explains a possible fixed-critic small-step benefit of substepping. It neither predicts the sign of true-return change nor applies to the projected, saturated, moving-critic branches studied at large T . The expansion is not a uniform $K \rightarrow \infty$ result.

5 EXPERIMENTS

We evaluate BAR on the nine D4RL locomotion tasks formed by HalfCheetah, Hopper, and Walker2d with medium, medium-replay, and expert datasets (Fu et al., 2020). These tasks directly exercise deterministic continuous control and provide distinct behavior-data regimes for a target-routing stress test. Base implementations follow CORL configurations (Tarasov et al., 2022); scores are D4RL normalized returns. Additional per-task results and diagnostic definitions are in the Supplementary Material.

Fixed-budget protocol and estimand. We hold a total nominal budget T fixed, split it across K actor hops as $h = T/K$, and sweep

$$T \in \{0.05, 0.1, 0.2, 0.4, 0.7, 1.5, 2.5, 4, 7, 10, 12, 14, 17, 20\}.$$

over all nine tasks with two seeds. The sweep is complete over this specified grid and contains $9 \times 14 \times 2$ runs per method. It is designed as a high-resolution stability phase diagram rather than a final benchmark alone. For $K = 1$, Proposition 1 maps T to the normalized TD3+BC coefficient $\alpha = 2T$.

We compare TD3+BC ($K = 1$), proximal BAR with $K \in \{2, 3, 4\}$ (BAR-Prox), and a scale-matched projected linearized target with $K \in \{2, 3\}$ (BAR-Lin):

$$a_{k+1}^{\text{tar}} = \Pi_{\mathcal{A}} \left[a_k + d \frac{T}{K} \frac{\nabla_a \hat{Q}(s, a_k)}{C_k} \right]. \quad (19)$$

Here a_0 is the sampled dataset action. At each hop, BAR-Lin detaches the current reference, recomputes the $\hat{Q}$ gradient and C_k , clips the target to the action box, and takes one mean-squared-error optimizer step with the corresponding persistent actor. It uses the same first-hop target and final-hop evaluation routing as BAR-Prox. The factor d matches the local action-space coefficient induced by the action-averaged metric in Eq. (6). All conditions match datasets, critic updates, and the horizon of 10^6 critic updates; checkpoint diagnostics use the final checkpoint. A K -hop condition performs K actor optimizer steps at each delayed policy update, and C_k is recomputed at each reference. The primary estimand is the end-to-end effect of the fixed-budget refinement bundle, which jointly changes local step size, re-centering frequency, actor compute, and target/deployment dataflow.

Table 1: Aggregate fixed-budget results. High denotes $T \geq 4$ and tail denotes $T \geq 10$. Collapse entries are seed-mean cells / raw seeded runs; P and L denote proximal and linearized.

Method	$T \leq 2.5$	High	Tail	Collapse
TD3+BC	68.35	25.97	20.46	35 / 68
BAR-P2	71.62	42.02	31.49	25 / 53
BAR-P3	70.56	52.78	43.55	17 / 41
BAR-P4	71.21	63.61	56.26	9 / 29
BAR-L2	69.02	36.68	27.44	29 / 58
BAR-L3	68.54	43.91	33.12	24 / 53

The analyses test whether this bundle reduces large-budget low-return outcomes, whether the effect persists under the linearized realization, how first-hop target-policy displacement differs from final-policy reach, and which critic-validity diagnostics characterize failed training trajectories.

5.1 Additional actors broaden the stability envelope

Figure 2 and Table 1 summarize the fixed-budget sweep. We descriptively call a high-budget cell collapsed when its two-seed mean score is below 20; this is a locomotion failure indicator, not a theoretical threshold. Across proximal $K = 1, 2, 3, 4$, the high-budget mean increases from 25.97 to 42.02, 52.78, and 63.61, while collapsed cells decrease from 35/63 to 25/63, 17/63, and 9/63. From $K = 3$ to $K = 4$, the mean cellwise gain is +10.83, but the median is only +.32: eight collapsed cells are rescued, none newly collapse, and most other cells change little. BAR-Lin with $K = 3$ also raises the high-budget mean from 36.68 to 43.91 and reduces collapsed cells from 29 to 24. Thus additional actors primarily rescue low-return outcomes rather than uniformly shifting settings. Threshold sensitivity and per-task means are in the Supplementary Material.

Task dependence and replication. The budget response is smooth on HalfCheetah-medium, plateau-like on Hopper-medium-replay, sharply nonmonotone on Hopper-medium and Walker2d-medium, and asymmetric on expert datasets. Within-task transitions are often organized more clearly by realized first-hop movement than by nominal T , but comparable movement can yield very different returns across tasks. There is therefore no universal critic-valid radius. A targeted four-seed Hopper-medium audit confirms that BAR-Prox $K = 4$ scores 100.9 ± 1.7 at $T = 17$, where $K = 3$ scores 0.7, but becomes highly variable at $T = 20$ (41.8 ± 44.0). The fourth actor relocates a nonmonotone high-return window; it does not estab-

lish a linear frontier law. Independently, the one-hop grid on seeds 2–3 has high-budget mean 24.79 and exactly 35/63 collapsed cells, closely reproducing 25.97 and 35/63 on seeds 0–1.

The linearized sweep shows that the gain is not unique to the proximal loss. At high budgets, BAR-Lin with $K = 3$ gains 7.23 points over $K = 2$ and reduces collapse cells from 29 to 24. BAR-Prox remains stronger overall, so these are comparisons of complete training procedures rather than isolated integrator effects.

5.2 Large-budget behavior is nonlocal and critic dependent

At $T = 20$, the raw linearized target reaches the action box in 94–95% of states, and the effective action-step coefficient h_{eff} is only 14% and 9% of the nominal dT/K coefficient in the two hops. These are medians over 18 environment–seed runs per hop. The practical operation is projected target generation followed by amortized neural regression under a moving critic. The sweep measures the resulting end-to-end stability envelope; local truncation error lies outside this estimand.

Figure 3 shows the common failure signature. Environment return supplies the external label: collapsed runs score below 20. These runs have orders-of-magnitude larger TD-error tails, larger critic magnitudes, and larger realized movements. The controlled common critic is the frozen final TD3+BC $T = 1$ critic matched by environment and seed. A run-total gain sums the mean Q change across actor hops on the shared state batch. For collapsed BAR-Lin runs, its median under the jointly trained critic is 3.33×10^{11} ; under the common critic it is -1.42 , compared with $+2.11$ for stable runs. For BAR-Prox, the common-critic gain changes from $+2.51$ in stable runs to -1.06 in collapsed runs. This sign reversal diagnoses cross-critic disagreement consistent with Eq. (14); environment return remains the criterion for policy failure.

A simulator-reference audit provides targeted evidence of critic miscalibration in the screened checkpoints: the median signed error $\hat{Q}_1(s, a_K) - Q_{\text{MC}}^{\rho_1}(s, a_K)$ is $+1.70 \times 10^{12}$ in five collapsed checkpoints and -38.7 in 55 stable ones. Its action-ranking subset is sparse, so we use it only as targeted checkpoint-level evidence.

5.3 Target-policy displacement and final-policy reach

At the same nominal budget, BAR-Prox realizes a smaller current-state first-hop displacement proxy than BAR-Lin once $T \geq 7$ (the Supplementary Material). At $T = 20$, their median $\hat{D}_{\text{critic}}$ values are 0.195

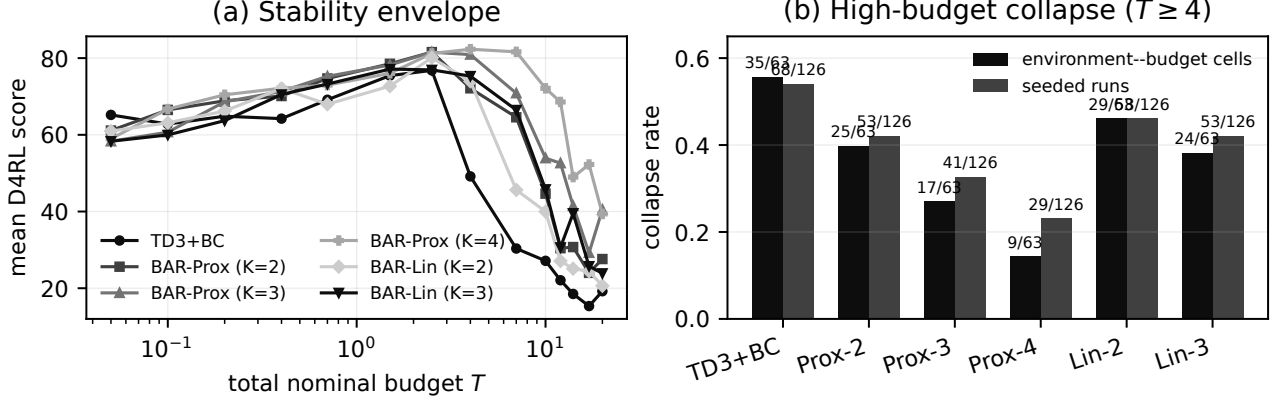


Figure 2: Fixed-budget sweep over nine locomotion tasks. (a) Mean score over tasks. (b) Collapse rates for $T \geq 4$, reported both by environment-budget cell and by seeded run. The tested refinement bundles broaden the stable high-budget region for both proximal and scale-matched linearized realizations.

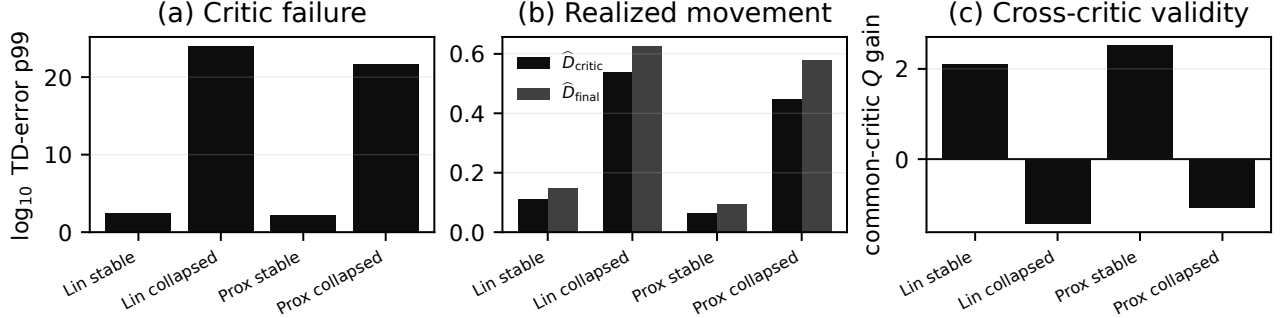


Figure 3: Median diagnostics over the matched $T \in \{4, 7, 10, 14, 20\}$ audit. Proximal and linearized realizations share a diagnostic pattern consistent with critic miscalibration: critic errors and realized movements increase sharply, and run-total gains under a common critic become negative. Environment return labels failure; the common critic diagnoses cross-critic disagreement.

and 0.323. Across 90 matched environment-budget-seed cells at $T \in \{4, 7, 10, 14, 20\}$, exactly one realization collapses in 11 cases. In all 11, the stable realization has both smaller $\hat{D}_{critic}$ and smaller $\hat{D}_{final}$. These matched-cell outcomes associate smaller realized movement with the stable optimization trajectory. Because the proxy is measured on current validation states rather than the Polyak target policy on next states, this result organizes the observed failure transition without identifying a causal route through the TD target.

We directly measure Eq. (12) on all 270 audited cells. The current-state proxy closely tracks deterministic target-policy displacement ($r = .9999$). BAR-Prox $K = 3$ has lower target-policy displacement than TD3+BC in 88 of 90 matched cells (median ratio .658), while the current-state $\hat{D}_{final}$ proxy is lower in only 36 of 90 (median ratio 1.057). Thus the three-hop

refinement can reduce movement in the target-policy branch without merely shrinking its measured final-policy reach. Against $K = 2$, $K = 3$ lowers target-policy displacement in 84 of 90 runs but loses by more than one score point in 17 of those 84, showing that target-policy displacement alone does not determine performance (the Supplementary Material).

A targeted $K = 3$ study on Hopper-medium reveals two modes of decoupling (the Supplementary Material). For $T = 4$ – 10 , $K = 2$ and $K = 3$ have nearly identical $\hat{D}_{critic}$, whereas $K = 3$ increases $\hat{D}_{final}$ and score; at $T = 10$, the scores are 58.12 and 85.35. In the observed transition region from stable to collapsed scores ($T = 12, 14$), $K = 3$ instead contracts both displacements and avoids the low-return outcome. At $T = 20$, both enter severe projection and saturation and collapse. The third hop changes target-policy movement, re-centering frequency, and the optimiza-

tion trajectory rather than the formal order of the local update.

5.4 Targeted component controls

We next hold the actor trajectory fixed and change only the backup route. Two actor-detached shadow critics start from the same parameters and receive identical minibatches and target-smoothing noise; one bootstraps through the first-hop target actor and the other through the final-hop target actor. The study covers Hopper-medium at $T \in \{10, 20\}$ and Walker2d-medium at $T \in \{4, 7\}$, with two seeds and four checkpoints. Relative to a common simulator return under the effective first-route backup continuation, the final-route shadow has greater state-median symmetric relative error in all eight run-level averages. The final-minus-first route delta has mean .124, median .013, a run-bootstrap mean 95% interval [.008, .323], and an exact two-sided sign-flip $p = .0078$. Because the final-route critic targets a different continuation, this contrast measures route-induced alignment with the primary first-route value estimand, not each shadow critic’s calibration to its own continuation. Neither shadow critic drives the actors, so the intervention does not estimate a routing effect on return.

Two additional eight-run controls probe simpler explanations (the Supplementary Material). With three actor optimizer calls in every condition, sequential $K = 3$, a fixed-reference chain, and a direct-final control have means 71.4, 70.0, and 59.4 and collapse counts 2/8, 2/8, and 3/8. Fixed-reference refinement therefore matches sequential re-centering in this targeted sample, so the experiment does not establish re-centering alone as the source of the sweep-wide gain. Slowing only the $K = 1$ actor-target Polyak rate by a factor of three changes the matched collapse count from 6/8 to 7/8 and does not reproduce the $K = 3$ rescue; actual displacement under this lag intervention was not audited. The route-only comparison characterizes alignment with the first-route estimand; together with the compute and lag controls, it refines the architectural diagnosis but does not identify routing as the cause of the sweep-wide gain.

5.5 Discussion and limitations

Both refinement realizations reduce large-budget failures, although BAR-Prox remains stronger overall. Direct measurements show that lower first-hop target movement need not shrink final-policy reach, and fixed-trajectory final-route shadows are less aligned with the common first-route value estimand in all eight targeted runs. Fixed-reference refinement matches sequential re-centering, while slower target averaging

does not rescue the selected one-hop cells. Thus the primary estimate remains the implemented bundle: K changes local step, re-centering, compute, normalization, routing, and the optimization trajectory together.

The cross- K grid covers nine locomotion datasets but only two random seeds per cell, so the reported means and collapse counts have limited seed-level precision. Collapse is descriptive, and the common-critic, simulator, and eight-run controls are targeted rather than exhaustive. We do not provide an offline rule for selecting T or K . The finite-step margin concerns the learned critic; the Wasserstein-flow account assumes small realized steps under a fixed critic.

6 CONCLUSION

BAR splits a fixed nominal actor budget across a persistent TD3+BC actor chain, coupling the critic target to the first branch and deployment to the last. Its squared anchors admit a deterministic Wasserstein proximal interpretation only in the fixed-critic local regime. In the complete two-seed grid, four proximal actors raise the high-budget mean from 25.97 to 63.61 and reduce low-return cells from 35/63 to 9/63; the near-zero median $K = 4$ improvement identifies the effect as lower-tail rescue. Three-hop BAR lowers directly measured target-policy displacement in 88/90 matched cells without generally shrinking the final-policy proxy. The evidence supports the end-to-end stability of the tested refinement bundle, not a causal effect of routing, re-centering, compute, or budget splitting alone.

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Checklist

1. For all models and algorithms presented, check if you include:
 - (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes/No/Not Applicable]
Yes. Sections 3–4 define the setting, actor chain, losses, and assumptions.
 - (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes/No/Not Applicable]
Yes. Section 4 states that BAR uses K persistent actors and K actor optimizer calls per delayed update; the theoretical properties are stated in Propositions 2–4.
 - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [Yes/No/Not Applicable]
Yes. The supplementary material specifies the implementation and dependencies; the
2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. [Yes/No/Not Applicable]
Yes. Each proposition states its assumptions, and the local-flow assumptions are separated from the coupled large-budget regime.
 - (b) Complete proofs of all theoretical results. [Yes/No/Not Applicable]
Yes. Complete proofs are provided in the supplementary material.
 - (c) Clear explanations of any assumptions. [Yes/No/Not Applicable]
Yes. Sections 4 and 5 explain the learned-critic, fixed- K , small-step, and implementation restrictions.
3. For all figures and tables that present empirical results, check if you include:
 - (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [Yes/No/Not Applicable]
Yes. Released CSVs, figure-generation code, a numerical claim verifier, and build instructions reproduce every main figure and aggregate table.
 - (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes/No/Not Applicable]
Yes. The fixed budget grid, datasets, seeds, update counts, actor schedule, evaluation protocol, and targeted-control protocols are specified in Section 5 and the supplementary material.
 - (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes/No/Not Applicable]
Yes. Captions and text define seed-mean cells, raw seeded runs, collapse thresholds, paired ratios, and the targeted four-seed mean and sample standard deviation. The main grid is explicitly identified as a two-seed descriptive phase diagram.
 - (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes/No/Not Applicable]
No. The released artifact records software and execution protocols but does not contain a complete, independently verified inventory of all external-machine hardware.

anonymized release contains launch, evaluation, diagnostic, and verification scripts.

4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
 - (a) Citations of the creator If your work uses existing assets. [Yes/No/Not Applicable]
Yes. D4RL, TD3+BC, and the CORL implementation basis are cited.
 - (b) The license information of the assets, if applicable. [Yes/No/Not Applicable]
Yes. The code release identifies its license; upstream dependencies retain their own licenses.
 - (c) New assets either in the supplemental material or as a URL, if applicable. [Yes/No/Not Applicable]
Yes. An anonymized code and compact-results artifact accompanies the submission.
 - (d) Information about consent from data providers/curators. [Yes/No/Not Applicable]
Not Applicable. We use public simulated-control benchmark datasets and collect no human data.
 - (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Yes/No/Not Applicable]
Not Applicable. The assets contain simulated locomotion trajectories and no personal or sensitive content.
5. If you used crowdsourcing or conducted research with human subjects, check if you include:
 - (a) The full text of instructions given to participants and screenshots. [Yes/No/Not Applicable]
Not Applicable.
 - (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Yes/No/Not Applicable]
Not Applicable.
 - (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Yes/No/Not Applicable]
Not Applicable.